# YOUR MACHINE LEARNING PROJECT TRACKER

8 hands-on projects | Supervised + Unsupervised + Deployment

**CHECK OFF EACH TASK AS YOU COMPLETE IT**

Student name: _____

Start date: _____

Target finish: _____

## PROJECT OVERVIEW

| | | |
|---|---|---|
| ☐ | **01 House Price Prediction** | 3-4 days |
| ☐ | **02 Bank Marketing Prediction** | 4-5 days |
| ☐ | **03 Wine Quality Prediction** | 4 days |
| ☐ | **04 Customer Segmentation** | 5 days |
| ☐ | **05 PCA and Cluster Visualization** | 3 days |
| ☐ | **06 Credit Card Fraud Detection** | 6-7 days |
| ☐ | **07 Customer Churn Prediction** | 6-7 days |
| ☐ | **08 End-to-End ML Prediction App** | 7-10 days |

## HOW TO USE THIS TRACKER

1. Open the PDF in a viewer that supports fillable checkboxes.
2. Mark tasks as you complete them and save a copy of your PDF.
3. Use each project page to record scores, blockers and mentor review.

PROJECT 01 / 08

# House Price Prediction

Supervised / Regression

**LEVEL** Beginner **ESTIMATED TIME** 3-4 days

**DATASET / RESOURCE**

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.fetch_california_housing.html

## TASK CHECKLIST

Done

- [ ] Load California Housing data into a Pandas DataFrame
- [ ] Display first 10 rows, shape, columns and dtypes
- [ ] Identify target and features (X and y)
- [ ] Check missing values and duplicates
- [ ] Create histograms and scatter plots
- [ ] Explore correlation with target; explain strongest relationship
- [ ] Split data into training and testing (80/20)
- [ ] Train Linear Regression baseline
- [ ] Train Decision Tree and Random Forest regressors
- [ ] Calculate MAE, RMSE and R2 on test data
- [ ] Compare models and describe overfitting risk
- [ ] Write README and publish code to GitHub

## REFLECTION / MENTOR REVIEW

- Rows / columns?
- Which column is the target?
- Missing values?
- Which feature correlates most with target?
- Why regression rather than classification?

Best result / metric: _____

Biggest lesson: _____



- [ ] **Project fully complete**

PROJECT 02 / 08

# Bank Marketing Prediction

Supervised / Classification

**LEVEL** Beginner - Intermediate **ESTIMATED TIME** 4-5 days

**DATASET / RESOURCE**

https://archive.ics.uci.edu/dataset/222/bank+marketing

## TASK CHECKLIST

Done

- [ ] Download the UCI Bank Marketing data
- [ ] Read data dictionary and define target label
- [ ] Check missing, unknown and inconsistent values
- [ ] Visualize class balance and major features
- [ ] Create preprocessing for categorical and numeric columns
- [ ] Use stratified train/test split
- [ ] Train DummyClassifier as a baseline
- [ ] Train Logistic Regression, Decision Tree and Random Forest
- [ ] Report precision, recall, F1 and confusion matrix
- [ ] Compare threshold choices and class imbalance impact
- [ ] Check for features unavailable at prediction time (leakage)
- [ ] Summarize business implications and publish project

## REFLECTION / MENTOR REVIEW

- When is recall more important than precision?
- Why can accuracy be misleading?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____ ☐ **Project fully complete**

PROJECT 03 / 08

# Wine Quality Prediction

Supervised / Classification + Regression

**LEVEL** Intermediate **ESTIMATED TIME** 4 days

**DATASET / RESOURCE**

https://archive.ics.uci.edu/dataset/186/wine+quality

## TASK CHECKLIST

Done

- [ ] Download red or white wine quality dataset
- [ ] Inspect target distribution and feature distributions
- [ ] Choose regression target or define a justified quality class
- [ ] Split train/test before data-driven preprocessing
- [ ] Scale features where needed using a pipeline
- [ ] Train at least three suitable models
- [ ] Evaluate with metrics matching problem type
- [ ] Use cross-validation on training data
- [ ] Tune hyperparameters without touching test set
- [ ] Compare feature importance or model coefficients
- [ ] Explain mispredictions and limitations
- [ ] Document model selection

## REFLECTION / MENTOR REVIEW

- Does scaling matter equally for every model?
- What metric fits your target choice?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____ ☐ **Project fully complete**

PROJECT 04 / 08

# Customer Segmentation

Unsupervised / Clustering

**LEVEL** Intermediate **ESTIMATED TIME** 5 days

**DATASET / RESOURCE**

https://archive.ics.uci.edu/dataset/352/online+retail

## TASK CHECKLIST

Done

- [ ] Download Online Retail transaction dataset
- [ ] Inspect cancellations, returns, null customer IDs
- [ ] Clean records and document assumptions
- [ ] Create customer-level RFM features
- [ ] Examine skew and transform features if necessary
- [ ] Scale clustering input variables
- [ ] Try K-Means with several cluster counts
- [ ] Compare elbow and silhouette methods
- [ ] Try hierarchical clustering or DBSCAN
- [ ] Profile clusters using interpretable business values
- [ ] Propose a distinct action for each segment
- [ ] Document cluster limitations and create visuals

## REFLECTION / MENTOR REVIEW

- How did you choose the number of clusters?
- Do clusters necessarily represent real categories?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____ ☐ **Project fully complete**

PROJECT 05 / 08

# PCA and Cluster Visualization

Unsupervised / Dimensionality Reduction

**LEVEL** Intermediate

**ESTIMATED TIME** 3 days

**DATASET / RESOURCE**

https://archive.ics.uci.edu/dataset/109/wine

## TASK CHECKLIST

Done

- [ ] Load the UCI Wine dataset
- [ ] Inspect the 13 numeric attributes
- [ ] Standardize variables before PCA
- [ ] Fit PCA without using labels as inputs
- [ ] Plot explained variance and cumulative variance
- [ ] Project data onto two principal components
- [ ] Visualize samples in 2D
- [ ] Apply K-Means and overlay cluster assignments
- [ ] Compare clusters to known classes for interpretation only
- [ ] Explain information lost when using two components
- [ ] Write findings and add visualizations to README

## REFLECTION / MENTOR REVIEW

- Why is scaling important before PCA?
- What does explained variance mean?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____

- [ ] **Project fully complete**

PROJECT 06 / 08

# Credit Card Fraud Detection

Supervised + Anomaly Detection

**LEVEL** Advanced

**ESTIMATED TIME** 6-7 days

**DATASET / RESOURCE**

https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

## TASK CHECKLIST

Done

- [ ] Download fraud transactions dataset
- [ ] Measure fraud rate and describe imbalance
- [ ] Create stratified holdout split
- [ ] Train simple classification baseline
- [ ] Train Random Forest or suitable supervised model
- [ ] Explore class weighting; resample training data only if needed
- [ ] Train Isolation Forest as an anomaly baseline
- [ ] Compare precision, recall, F1 and PR-AUC
- [ ] Tune decision threshold on validation data
- [ ] Review false positives and false negatives
- [ ] Evaluate once on held-out test set
- [ ] Write ethical and operational limitations

## REFLECTION / MENTOR REVIEW

- Why is PR-AUC useful for rare fraud?
- What happens when fraud is missed?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____ ☐ **Project fully complete**

PROJECT 07 / 08

# Customer Churn Prediction

Supervised / Business Classification

**LEVEL** Advanced **ESTIMATED TIME** 6-7 days

**DATASET / RESOURCE**

https://www.kaggle.com/datasets/blastchar/telco-customer-churn

## TASK CHECKLIST

Done

- [ ] Download Telco Churn dataset
- [ ] Define churn label and business objective
- [ ] Clean blanks and convert numeric columns
- [ ] Explore churn by contract, tenure and charges
- [ ] Build preprocessing + model pipelines
- [ ] Create baseline and train two stronger models
- [ ] Compare F1, recall, precision and ROC-AUC
- [ ] Tune a decision threshold for a chosen business goal
- [ ] Explain predictions with feature effects
- [ ] Propose targeted retention actions
- [ ] Save model and preprocessing pipeline
- [ ] Present findings and add a strong GitHub README

## REFLECTION / MENTOR REVIEW

- Why does churn recall matter?
- What are risks of interpreting correlation causally?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____ ☐ **Project fully complete**

PROJECT 08 / 08

# End-to-End ML Prediction App

ML Engineering / Deployment

**LEVEL** Advanced **ESTIMATED TIME** 7-10 days

**DATASET / RESOURCE**

Use the best model from one of the projects above

## TASK CHECKLIST

Done

- [ ] Select a tested model and define its inputs
- [ ] Refactor notebook into reusable Python modules
- [ ] Build a reproducible training script
- [ ] Serialize model and preprocessing safely
- [ ] Create FastAPI /predict endpoint with validation
- [ ] Write several unit tests for prediction inputs
- [ ] Create a simple Streamlit interface
- [ ] Add requirements file and setup instructions
- [ ] Containerize the service with Docker
- [ ] Test locally from a fresh environment
- [ ] Document model limitations and error handling
- [ ] Push code with screenshots and usage examples to GitHub

## REFLECTION / MENTOR REVIEW

- Can a stranger run your app using README alone?
- What happens if a user enters an invalid value?

Best result / metric: _____

Biggest lesson: _____

Completed on: _____ ☐ **Project fully complete**